



Sequence Labeling in NLP

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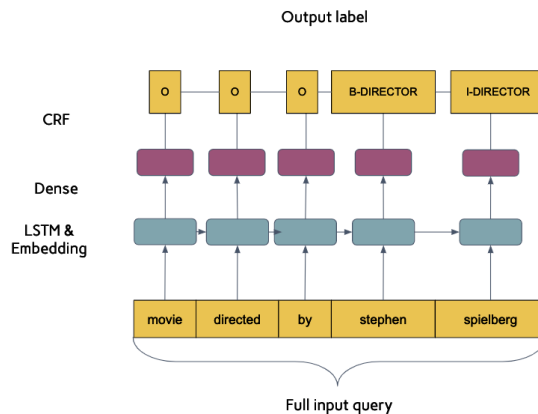
Objectives

- Sequence Labeling : Token-level Classification
- Part of Speech: English Word Classes
- Part of Speech Tagging
- Named Entities and Named Entity Recognition
- Evaluation



Sequence Labeling

- An important problem in the NLP pipeline. A lot of NLP problems are modelled as sequence labeling tasks : POS Tagging, Named Entity Recognition
- Important preprocessing step for many NLP pipelines. Tasks like NER can be precursor to a lot of pipelines running at the background of Language AI
- Problem approach: Token Classification
 - View text as a sequence of words/token
 - Every token in the sequence is assigned an individual label
 - Once labels are assigned to all tokens, they are decoded to obtain outputs from the entire label
 - Label for individual tokens derive dependencies from the entire sequence
- General framework for implementing Sequence Labeling (and other NLP tasks):
 - Train a language model for optimal embedding generation for token
 - Extend LM with a neural network that sits on top to use generated embeddings and generate label for downstream task – sequence-level outputs



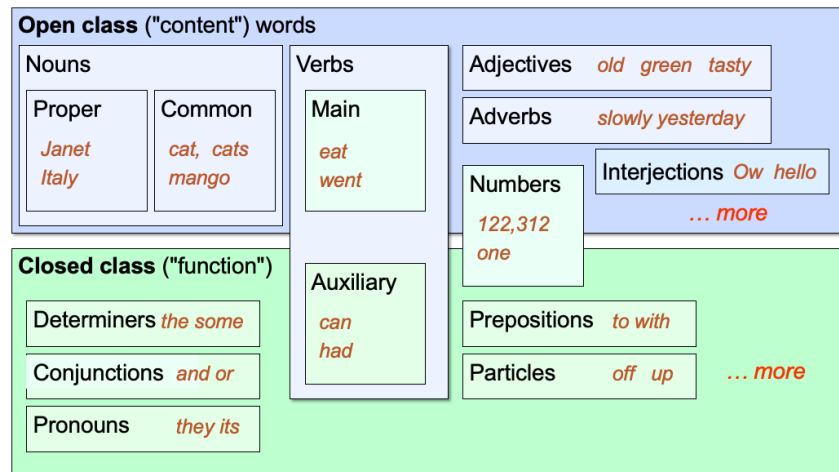
POS Tagging

The idea that words can be classified into grammatical categories. While these word classes do have semantic tendencies—adjectives, for example, often describe *properties* and nouns *people*— parts of speech are defined instead based on their grammatical relationship with neighboring words or the morphological properties about their affixes.

8 commonly identified parts of speech: noun, verb, pronoun, preposition, adverb, conjunction, participle, article

Two classes of words for Parts of Speech : Open vs Closed

- Open class words – membership to this class is always evolving, not fixed
 - Usually content words: Nouns, Verbs, Adjectives, Adverbs
 - Plus interjections: oh, ouch, uh-huh, yes, hello
 - New nouns and verbs like iPhone or to fax
- Closed class words – Relatively fixed membership
 - Usually function words: short, frequent words with grammatical function
 - determiners: a, an, the
 - pronouns: she, he, I
 - prepositions: on, under, over, near, by, ...



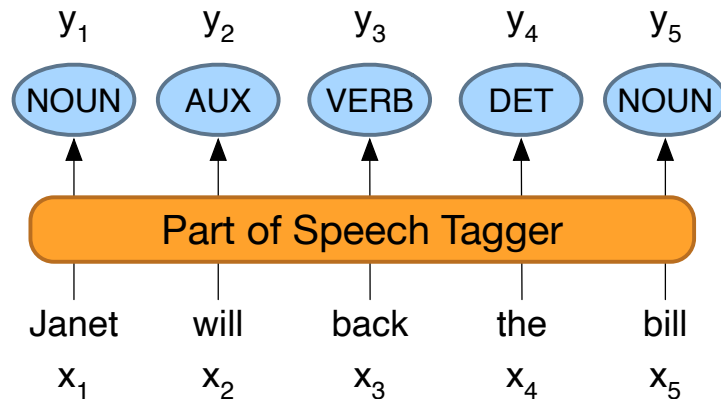
Parts-of-Speech Tagging

- Assigning a part-of-speech tag as label to each word in a text.
- Tagging is a **disambiguation** task – Words can have more than one POS. The goal is to find the correct POS tag for the immediate context.
 - VERB: (Book that flight)
 - NOUN: (Hand me that book)

Problem: Here's a sketch of part of speech tagging. The input is a sequence $x_1, x_2, \dots, x_n$ of (tokenized) words and a **tagset**, and the output is a sequence $y_1, y_2, \dots, y_n$ of tags, each output y_i corresponding exactly to one input x_i . Map from sequence $x_1, \dots, x_n$ of words to $y_1, \dots, y_n$ of POS tags

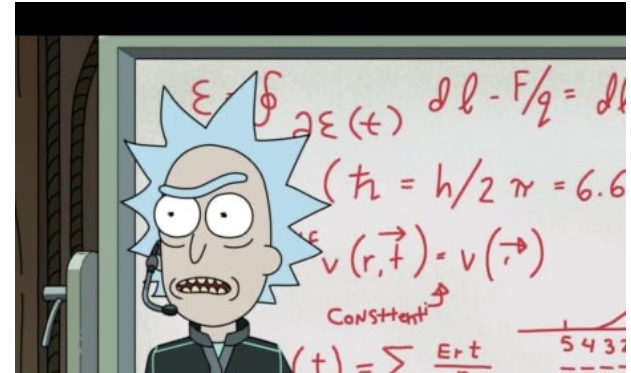
Sample “Tagged” English Sentences: Here’s what outputs of part of speech tagging will look like.

- There/**PRO** were/**VERB** 70/**NUM** children/**NOUN** there/**ADV** ./**PUNC**
- Preliminary/**ADJ** findings/**NOUN** were/**AUX** reported/**VERB** in/**ADP** today/**NOUN** 's/**PART** New/**PROPN** England/**PROPN** Journal/**PROPN** of/**ADP** Medicine/**PROPN**



How difficult is POS Tagging in English?

- Roughly 15% of word types are ambiguous
 - Hence 85% of word types are unambiguous
 - *Janet* is always PROP, *hesitantly* is always ADV
- But those 15% tend to be very common. So ~60% of word tokens are ambiguous
 - E.g., *back*
earnings growth took a *back*/ADJ seat
a small building in the *back*/NOUN
a clear majority of senators *back*/VERB the bill
enable the country to buy *back*/PART debt
I was twenty-one *back*/ADV then
- How accurate are modern POS taggers?
 - Quite accurate; for English, and a for a few other languages where we have sufficient hand-labeled training data, and which have sufficiently simple morphology, tagging is a mostly solved problem.
 - For English: The accuracy, that is, the percentage of tokens that the tagger assigns the correct tag to, is about 92% for Baseline (non-neural) systems and upto 97% for more sophisticated models.



Named Entities

Named entity, in its core usage, means anything that can be referred to with a proper name. Most common 3 tags:

- PER (Person) : Marie Curie
- LOC (Location) : New York City
- ORG (Organization) : University of Chicago
- Often multi-word phrases.
- Also, the term named entity has a loose definition. Often extended to dates, times, prices etc. that are a specific concept but not always proper nouns.

The task of named entity recognition (NER) :

- find spans of text that constitute proper names
- tag the type of the entity.

Why NER?

- Sentiment analysis: consumer's sentiment toward a particular company or person?
- Question Answering: answer questions about an entity?
- Information Extraction: Extracting facts about entities from text.

Albert Einstein PER Albert Einstein was born in Ulm LOC in
Germany LOC on March 14, 1879. Six weeks later the family moved to
Munich LOC , where he later on began his schooling at the Luitpold
Gymnasium ORG . In 1896 he entered the Swiss Federal Polytechnic
School ORG in Zurich LOC to be trained as a teacher in physics and
mathematics.

Named entity recognition outputs for PER, LOC and ORG

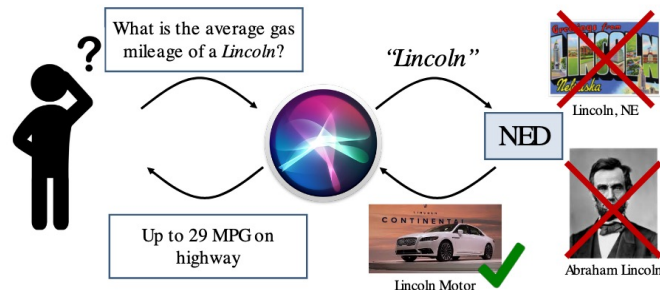
Why is NER hard?

NER is not a straight-forward token classification problem.

- Segmentation
 - In POS Tagging, no segmentation problem since each word gets one tag, however not every token/word in a sentence is associated with a named entity.
 - In NER we must find and segment the sentence into entities and non-entity tokens.
- Type Ambiguity
 - Surface forms of entities can refer to different types depending on the context it references
 - The entity types to be considered for labeling are fixed at the start of the problem.

[PER Washington] was born into slavery on the farm of James Burroughs.
[ORG Washington] went up 2 games to 1 in the four-game series.
Blair arrived in [LOC Washington] for what may well be his last state visit.
In June, [GPE Washington] passed a primary seatbelt law.

The name Washington can refer to the educator Booker T Washington, the sports team, the location, or the geo-political entity.



BIO+ Encoding for unified NER solution

NER is a structured two-pronged problem – Segmentation + Typing. How to turn it into a regular sequence labeling problem like POS Tagging with one label per word?

[**PER** Jane Villanueva] of [**ORG** United] , a unit of [**ORG** United Airlines Holding] , said the fare applies to the [**LOC** Chicago] route.

Solution: for a span-recognition problem like NER adopt a form of BIO tagging (Ramshaw and Marcus, 1995). Treat NER like a token classification task, via **tags that capture both the location of the token w.r.t. the nearest entity boundary and the type of that named entity**.

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

Different types of BIO Encoding

Standard Algorithms for NER

Supervised Machine Learning given a human-labeled training set of text annotated with tags

- Hidden Markov Models
- Conditional Random Fields (CRF)/ Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers)
- Large Language Models (like BERT), finetuned

We will mainly focus on the last two models!

